

1. Administrative & Study Identification

Full Study Title: Durvalumab and Tremelimumab With Lenvatinib as First-line Treatment in Patients With Unresectable Hepatocellular Carcinoma (TREMENDOUS-2) **Short Title/Acronym:** TREMENDOUS-2 **Protocol Number:** D419CL00025 **NCT Number:** NCT07081633 **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Durvalumab, Tremelimumab, and Lenvatinib **Phase of Development:** Phase II **Trial Status:** Recruiting (Currently recruiting participants) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the efficacy of durvalumab and tremelimumab in combination with lenvatinib as measured by Progression Free Survival (PFS) in participants with unresectable hepatocellular carcinoma (HCC). **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), safety (including Treatment-Related Adverse Events), and patient-reported outcomes.

2.2 Background & Rationale To evaluate the clinical benefit and tolerability of combining dual immune checkpoint inhibition (durvalumab and tremelimumab) with a multi-kinase inhibitor (lenvatinib) as a first-line treatment. This study addresses the unmet medical need to improve progression-free survival and overall response rates in patients with unresectable HCC who have not received prior systemic therapy.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Single-arm) **Status:** Recruiting

3.2 Planned Enrollment & Duration **Target N\$:** 114 patients **Number of Sites:** Approximately 24 locations **Estimated Study Start:** 05-Aug-2025 **Estimated Primary Completion:** 14-Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Confirmed HCC based on histopathological findings from tumor tissues or radiologically findings. 3. Must not have received prior systemic therapy for unresectable HCC. 4. Barcelona Clinic Liver Cancer

(BCLC) stage B (that is not eligible for locoregional therapy) or stage C. 5. Child-Pugh Score class A. 6. ECOG performance status of 0 or 1 at enrollment. 7. At least 1 measurable lesion per RECIST 1.1 guidelines.

4.2 Exclusion Criteria 1. Any unresolved toxicity National Cancer Institute (NCI) Common Terminology Criteria for Adverse Event (CTCAE) Grade ≥ 2 from previous anticancer therapy. 2. History of hepatic encephalopathy within past 12 months or requirement for medications to prevent or control encephalopathy. 3. Clinically meaningful ascites. 4. Patients with main portal vein thrombosis. 5. Active or prior documented GI bleeding. 6. Patient currently exhibits symptomatic or uncontrolled hypertension. 7. Patients co-infected with HBV and HCV, or co-infected with HBV and hepatitis D virus (HDV). 8. Uncontrolled intercurrent illness.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Investigational Products: Durvalumab, Tremelimumab (Trade Name: Imjudo, ATC Code: L01FX20), and Lenvatinib. **Dosage/Frequency:** Durvalumab and Tremelimumab administered at protocol-defined dosing intervals, in combination with daily Lenvatinib. **Route of Administration:** Intravenous (IV) for Durvalumab and Tremelimumab; Oral for Lenvatinib. **Duration of Treatment:** Treatment continues until objective disease progression, unacceptable toxicity, withdrawal of consent, or other discontinuation criteria are met.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for HCC and management of symptomatic adverse events. **Prohibited:** Other concurrent systemic anti-cancer therapies, locoregional liver therapies, and investigational agents during the treatment phase.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging (CT/MRI), Baseline Labs, ECOG
Baseline	Day 0	Enrollment confirmation, First Dose of combination therapy, Safety Audit
Interim	Treatment Cycles	Safety Labs, Efficacy Assessment (Tumor Scans), irAE Evaluation
End of Study	At Progression	Final Vital Signs, Exit Interview, IP Accountability, Follow-up for Survival

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression Free Survival (PFS) per RECIST 1.1—measured from the date of the first dose until the

date of objective disease progression per RECIST 1.1 or death, whichever comes first. To be assessed when approximately 69 PFS events have occurred (approx. 60% maturity). **Measurement Method:** Investigator assessment and radiological evaluation using CT or MRI scans.

7.2 Secondary & Exploratory Endpoints 1. Grade ≥ 3 Treatment-Related Adverse Events (TRAE) within 6 months after the initiation of study intervention. 2. Overall Survival (OS). 3. Objective Response Rate (ORR) per RECIST 1.1. 4. Patient-reported outcomes.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring by Investigator; events graded using NCI CTCAE. Special attention is placed on immune-related AEs (irAEs) and lenvatinib-specific toxicities (e.g., hypertension, GI bleeding). **Serious Adverse Events (SAEs):** Must be reported to the sponsor/pharmacovigilance team within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent safety review board will evaluate patient safety data at regular intervals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary and secondary efficacy analyses. Safety Set for all subjects who receive at least one dose of study treatment. **Sample Size Justification:** Target sample size of $N=114$ is adequately powered to assess the primary endpoint, driven by observing approximately 69 PFS events (60% maturity) expected roughly 8 months after the last patient is dosed. **Handling of Missing Data:** Time-to-event outcomes (PFS, OS) will use standard right-censoring methods (Kaplan-Meier estimates) for patients without events or lost to follow-up.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits by Clinical Research Associates (CRAs) to ensure data integrity and protocol adherence. **Audit Trail:** eCRF locking, source data verification, and secure data clarification forms.

11. Study Identifiers & Governance

Primary ID: D419CL00025 **Registry Number:** NCT07081633 **Posting ID:** 1138203022013343659 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** TREMENDOUS-2 **Official Regulatory Title:** An Open-label, Multi-center Phase II Study of Durvalumab and Tremelimumab With Lenvatinib as First-line Treatment in Patients With Unresectable Hepatocellular Carcinoma

12. Clinical Framework (Oncology Specific)

Primary Condition: Unresectable Hepatocellular Carcinoma (HCC) **Diagnosis Threshold:** Histopathologically or radiologically confirmed HCC (BCLC Stage B/C) not eligible for locoregional therapy **Medical Methodology:** Triple combination systemic therapy (Dual Immuno-oncology + Tyrosine Kinase Inhibitor) **Optimization Target:** Maximize progression-free survival (PFS) and overall survival (OS) in a first-line setting

13. Study Procedures & Methodology

Management Pathway: Standard oncologic pathway for advanced HCC **Education Protocol (HCP):** Site initiation visits and continuous protocol training regarding immune-related adverse event (irAE) and TKI toxicity management. **Education Protocol (Patient):** Education on oral treatment adherence (lenvatinib) and prompt reporting of side effects (e.g., bleeding, blood pressure changes). **Quality Audit Mechanism:** Routine source data verification and central/independent radiological review of tumor scans.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continues until objective disease progression or death **Onsite Visit Cadence:** Coincides with treatment administration cycles (e.g., Q3W or Q4W per protocol). **Remote Monitoring Frequency:** Continuous pharmacovigilance and query resolutions. **Checklist Review Interval:** Tumor assessments conducted every 8 to 12 weeks to monitor for progression per RECIST 1.1.

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator				[Insert Name]			____		[DD-MMM-YYYY]							
	Clinical Operations Lead				[Insert Name]			____		[DD-MMM-YYYY]							
	Lead Statistician				[Insert Name]			____		[DD-MMM-YYYY]							